

Throwaway societies: a field survey of the quantity, nature and distribution of litter in New Jersey

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Abstract

This paper examines the distribution, composition and accumulation of litter in New Jersey. Using a geographical information system, litter was collected, counted and classified for 37 representative sites throughout the state. A total of 9989 pieces of litter were counted over the four-week sampling period. The composition of litter was dominated by paper and plastic packaging when measured by area covered by litter, and by cigarette butts when measured by piece count. Accumulation rates varied according to land use, with urban residential and commercial sites having the highest levels of accumulation. Of five specialized environments (roadways, beaches, waterways, state parks and landfills), roadways have the largest area covered by litter, but litter accumulates more quickly in state parks.

The nature of the problem

Solid waste generation and disposal is a looming environmental problem in the US that is reaching crisis proportions in many regions. The US currently has one of the highest annual per capita solid waste generation rates in the world (762 kg). Similar industrialized countries produce far less waste. Japan, for example, has an annual municipal solid waste generation rate of only 344 kg per capita. European countries have similar annual rates: the UK is 313 kg per capita, Spain is 275 kg per capita, Germany is 317 kg per capita and Norway is 474 kg per capita. Only Australia (679 kg per capita) and New Zealand (656 kg per capita) come close to the US rate (World Resources Institute 1990).

The sheer volume of solid waste generated on an annual basis in the US poses major environmental challenges not yet encountered in the rest of the world. Landfills, the preferred disposal option, are reaching capacity in many states, and new ones are virtually impossible to site because of public opposition (US Congress 1989). Incinerators are problematic because of siting problems and air quality concerns. Recycling is one alternative for waste reduction which is having limited success in some areas, although there is still a need for disposal options (Blumberg and Gottlieb 1989; Newsday 1989). Clearly, America's solid waste crisis is forcing a change in consumer behaviour and a rethinking of the 'throwaway ethos'.

One byproduct of the American consumer society is the propensity to wantonly discard post-consumer waste wherever it is most convenient (along roads, in the street, in the local parks, and sometimes in rubbish bins). Litter is the name given to such material and is defined as any used or unconsumed substance or waste material that has been discarded (including beverage containers, paper wrappings, cigarettes, cigars and matches).

This paper demonstrates the utility of geographical methods in analysing litter accumulation by focusing on the results of a statewide litter survey conducted during the summer of 1989 on behalf of the New Jersey Department of Environmental Protection (NJDEP) (Cutter *et al.* 1990). Specifically, the paper examines the spatial distribution, composition and accumulation of litter and provides an assessment of the nature of the litter problem in New Jersey as it relates to land use.

Extensive litter reduction programmes have been in place for years with such catchy slogans such as 'Give a hoot, don't pollute', and 'Put litter in its place', the anthem of the Keep America Beautiful organization. Despite these educational efforts, litter is still a pervasive problem throughout the US and one of the least studied aspects of solid waste management in the country.

A review of the relevant literature on litter indicated the lack of geographical perspectives on litter. For instance, there is very little work on the spatial distribution of litter, or its composition and rate of accumulation, let alone discussions on variations from place to place. As a topic for geographical analysis, litter has been completely overlooked by the research community. There are some general studies on the litter problem, primarily pamphlets by the Keep America Beautiful organization (Finnie 1985), but no systematic analysis of litter or its geographic extent. More than 50 actual litter surveys have been conducted in the US since 1970, of which 17 were statewide surveys conducted by state personnel (usually highway departments) and the remainder were done by a single consulting firm specializing in this field (Syrek 1986).

Why people discard litter, or engage in littering behaviour, is the topic given most attention by academic researchers. Empirical studies of littering behaviour appear regularly in the environmental psychology literature, focusing mainly on littering in both recreational and urban environments (Robinson 1976). Behaviour modification research (Bacon-Prue *et al.* 1980; O'Neill *et al.* 1980; Reiter and Samuel 1980) and the use of a behavioural approach to litter management are also prevalent themes (Geller *et al.* 1976; Geller 1980, 1985; Geller and Lehman 1986; Roalesnieto 1988). Finally, there are a few studies on the effectiveness of legislation, primarily bottle bills, on changing littering behaviour (Levitt and Leventhal 1986; Lee *et al.* 1988).

Methods of measuring litter

Although there is no standard method (Sullivan 1985), the literature identifies two major methods for measuring litter and conducting litter surveys: item-orientated or full-scale surveys, and visually-orientated surveys (Marquez and Zandi 1985; Syrek 1985, 1986). Item-orientated surveys require that litter in specified places be counted and classified. This approach generates information on the composition and spatial impact of litter. Visual surveys, on the other hand, generate data on a site's visual appearance based on the number of items counted. In this type of survey, litter is either photographed or simply counted by a surveyor moving at a specified speed. Visual surveys are useful in determining a site's aesthetic appearance (defined by the number of littered items), but do not provide data on the type of litter or its areal coverage. In addition, visual surveys tend to underestimate the smaller items of litter, yet these pose one of the most difficult problems for a litter-reduction programme.

The most recognized method for conducting litter surveys is that of Syrek (1986), which measures accumulated and generated litter. One of the major difficulties with litter measurements is extrapolating statewide data from a limited sample of site types: Syrek's work embodies this problem, and his methodology for site selection and subsequent data extrapolation is never fully explained. Instead, Syrek developed

several statistical manipulations or 'reduction factors' (as he labels them) to control for differences in ground cover, traffic volume, pedestrian volume, weather conditions, median household income and other variables. Unfortunately, the use of such rigid reduction techniques can mask statistically significant differences (in site size and land-use type, for example) between sites within a state, as well as between different states. Thus, our ability to compare the distribution of litter from one state to another, or between cities, is extremely limited. Furthermore, the sites that are often selected may not be representative of the range of land uses in the state, again clearly diminishing the possibility of making comparisons across and between states, cities or regions. Methodologically, then, there are major problems with Syrek's sampling units and their representativeness.

Since there is so little work other than Syrek's concerning the methodology of litter surveys, critiques and alternatives are few. However, Marquez and Zandi (1985) compared the data obtained through full-scale and visual surveys. They found that while resulting litter counts vary widely, the overall picture—the rankings of sites and states in terms of litter inundation—generally hold constant, regardless of the methodology used. The primary difference between the methods is the quality and quantity of data collected, including the ability to compare different sites, an important consideration for the geographer.

This current study combines aspects of both full-scale and visual litter surveys. The data-collection phase of the project included multiple, item-orientated collection and sorting of litter, which provided data on litter composition, areal coverage and rates of fresh litter accumulation. Photographs taken during each site visit also revealed the degree to which larger pieces of litter pose aesthetic problems, since smaller items are usually less visible.

Site selection in New Jersey

Because of the methodological problems of sampling in Syrek's work, great care was taken to develop a sampling framework that included a standard range of land uses and more specialized environments such as beaches, roadways and landfills. Communities were selected using a stratified random site-selection methodology in order to derive a representative sample of communities within New Jersey. Thus, it was possible to aggregate data by community type (urban, rural, suburban, or residential, commercial, industrial) in order to assess relative differences between them.

A typology of New Jersey's 567 municipalities was developed using 17 community characteristics (Table 1). A principal components analysis (PCA) was used to categorize communities with respect to the 17 variables, paralleling similar community typology studies of New Jersey (Lake and Cutter 1980; Airola and Parker 1983). Seven community types were identified:

1. large, populated municipalities participating in solid-waste-recycling programmes;
2. cities with low-income minority residents;
3. communities with affluent white residents;
4. rapidly growing communities;
5. urban, densely populated communities, as opposed to rural, sparsely populated communities;
6. industry-based communities;

Table 1. Variables used in the classification of communities*Population related*

1. Population density
2. Population change 1980–7
3. Percentage population change 1980–7
4. Percentage minority (African-American, Spanish-speaking) population 1980
5. Percentage of municipal solid waste recycled
6. Per capita income index

Traffic related

7. Miles of federally funded roadways
8. Miles of state roadways
9. Miles of county roadways
10. Miles of municipal roadways
11. Density of roadways (miles of roadway mile⁻²)

Land-use related

12. Percentage of municipal land vacant (percentage of total real estate tax valuation classified as vacant land)
13. Percentage of municipal land industrial
14. Percentage of municipal land commercial
15. Percentage of municipal land farmland
16. Percentage of municipal land single-home residential
17. Percentage of municipal land apartment houses

7. communities with major commercial developments and a disproportionately large number of federal and state road miles.

Those communities best representing each category were then selected based on their individual category score (component scores $> +1.0$ or < -1.0) and locational criteria representing northern, central and southern New Jersey to ensure an even regional coverage. Twenty-one communities (seven from each region) were selected in this manner (Fig. 1).

Areas representing special types of land uses of concern to the funding agency (NJDEP) were added. These included:

1. communities with interstate highways and toll roads;
2. communities with many state and federal highway miles;
3. rural communities with county-controlled roadways;
4. communities with state park lands;
5. communities with beaches (both highly developed and less developed);
6. communities with rivers;
7. communities near solid waste landfills.

The selection criteria for these 'specialized locales' were less rigorous. For example, roads (county, state and toll/interstate) were selected on the basis of traffic volume and location (north, central, south). Park sites were based on high visitor use and location. Beach communities were chosen to represent northern and southern beaches, but also to ensure that bay beaches as well as ocean beaches were included. Waterways were selected to account for location (northern versus southern) and surrounding population density (urban and suburban locales). Finally, landfill sites were selected based on the number of trucks per day and represent high-volume and low-volume sites. An additional 16 sites were selected in this manner to produce a total of 37 (Table 2 and Fig. 1).

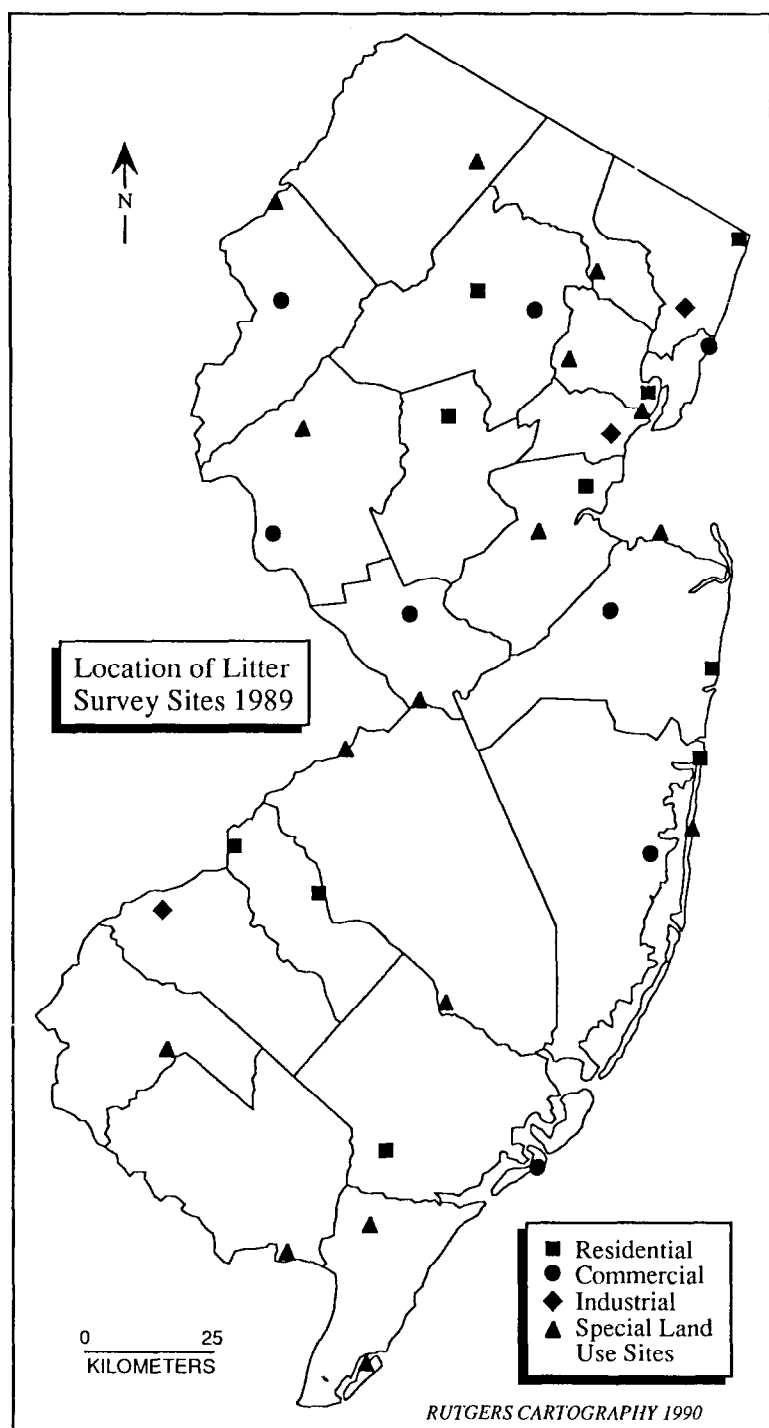


Figure 1. The location of the 37 sampling sites for the litter survey. Communities (residential, commercial, industrial) are differentiated from special land-use sites (beaches, roadways, watercourses, state parks and landfills).

Table 2. Survey sites classified by land use

Category	Site
Residential	
Urban	Asbury Park, Camden, Newark
Suburban	Alpine, Far Hills, Mantoloking, Victory Gardens, Woodbridge
Rural	Estell Manor, Voorhees
Commercial	
Urban	Atlantic City, West New York
Suburban	Berkeley Township, Lawrence Township, Marlboro Township, Parsippany-Troy Hills
Rural	Hope, Kingwood Township
Industrial	Greenwich Township, Linden, Moonachie
Roads	
Rural/county	Alloway Township, Hardyston Township
State/federal highway	Bordentown Township, Livingston
Interstate/toll road	Interstate 78 (Union Township), New Jersey Turnpike (Elizabeth)
Beaches	
Undeveloped bay	Fortescue (Downe Township)
Developed bay	Keansburg
Developed ocean	Seaside Heights, Wildwood
Watercourses	Delaware River (Burlington City), Passaic River (Wayne Township)
State parks	Wharton State Forest (Batsto), Worthington State Forest (Pahaquarry Township)
Landfills	Edgeboro Landfill (East Brunswick Township), Cape May County Landfill (Woodbine)

Once the 37 host communities had been selected, specific sites within each one were selected based on an initial field reconnaissance in each community. Each site had to meet a number of criteria:

1. It had to be accessible by foot, unencumbered by ditches, tall grass, marshlands, or other potentially dangerous conditions.
2. It had to be publicly owned (public rights of way) or at least publicly used (nearby parking lot areas).
3. It had to be safe—not too close to heavy vehicle traffic or hidden behind blind turns in a road.
4. It had to be large enough to accommodate a study plot (500 ft² or 46·5 m²).
5. It had to reflect the original classification (a suburban site for a suburban community, a beach site for a beach community, and so on).

Field methods and procedures for establishing the GIS in New Jersey

Sampling was done at two-week intervals over a four-week period during the summer of 1989. Each site was visited three times. To facilitate data collection and entry, a geographic information system (GIS) was established at each site. The site was divided into grids, each one composed of 20 cells (5 × 5 ft or 1·5 × 1·5 m). The geometry of

the grid varied from site to site to allow for local conditions, but most grids were parallel to the contours of roadways or pavements and were either rectangular or square. The grids were marked with pink flags and mapped with reference to a local datum point. This was necessary so that subsequent site visits could follow the same grid layout. The grid approach was applied here for four reasons:

1. It resulted in more organized, manageable, and standardized data collection within sites.
2. It allowed for systematic photography as a source of visual data.
3. It allowed for statistical comparisons between sites, since it standardized site size even when site geometry varied.
4. It allowed for the creation of computer-generated maps showing tendencies for litter cluster within sites (that is, GIS).

Each visit involved several tasks. First, field teams photographed the surroundings of each site to keep a record of local conditions. Secondly, after marking each of the 20 cells in a site, the field team photographed every square. Then the litter collection began. For each cell, accumulated litter was counted, catalogued and removed. Only pieces of litter larger than one inch (2.5 cm) in any dimension were included in this study, although small pieces of broken glass clustered in one spot were recorded as an instance of 'broken glass'. The first visit counted all litter up to that point in time. The second and third visits took place at weeks 2 and 4 and measured the accumulation of litter from the previous visit. At the end of the sampling period, a representative sample of collected litter was returned to the laboratory for size measurement in order to calculate the areal coverage of litter by major category such as paper, glass, plastic, styrofoam or metal. It proved impossible to calculate areal coverage from the photos because of the small size of most of the litter and the lack of visual contrast in the photos.

The total area sampled combining all 37 sites for all three visits was approximately 1.3 acres (0.53 ha), which represents only a very small fraction (0.003 per cent) of the total public land in the state. Additionally, the total area covered with litter was quite small (675 ft² or 64 m²), or about 1.2 per cent of the total area sampled. Because of the small area sampled, generalizing the results to determine the total amount of litter in the entire state is dangerous. However, data can be provided on accumulation rates and composition among different types of land uses throughout the state. This comparative measure further allows the development of targeted policies for litter reduction for a variety of land uses in the state.

Results

The results of the survey are presented in two ways. First, a summary of all data collected during all site visits provides information on the total amount of litter collected, its composition, and the weekly rate of fresh litter accumulation. Secondly, comparisons are made among different types of sites in the survey (rural, urban and suburban) to illustrate the spatial variability in litter accumulation and composition.

Litter composition

In three visits to 37 sites, the field teams collected a total of 9989 pieces of litter, with almost half the litter being collected during the first visit. The composition may be slightly biased toward non-degradable items since the time frame of accumulation prior to the survey is unknown. By actual item count, the single most frequent item

was the cigarette butt, accounting for almost a quarter of all pieces collected (Table 3). A total of 2321 butts (the equivalent of more than 116 packs of smoked cigarettes) were counted. Smoking-related (butts, matches and cigarette packages) items made up 28 per cent of the litter. The next most dominant type of litter was paper packaging (19 per cent), followed by plastic packaging (10 per cent), unmarked and thus unidentifiable pieces of paper (10 per cent), broken pieces of glass (9.5 per cent) and styrofoam (8 per cent).

In addition to the item counts, it was possible to estimate the amount of area covered by litter. Less than 2 per cent of our sampled area had litter on it. Therefore, while the individual item counts may seem high, the actual amount of land surface covered with litter is relatively small. While cigarette butts are the most pervasive litter in total item counts, their areal coverage is quite small (less than 0.5 per cent of total littered area) (Table 3). Those items with the greatest areal coverage include plastic packaging (35 per cent) and paper packaging (29 per cent). In terms of relative importance, paper and plastic litter cover about 81 per cent of the littered area in the sample.

By classifying litter according to its source, such as beverage containers or fast-food items, the amount of deliberate litter (wilfully discarded by individuals) that ends up in the environment can be estimated. The largest category of deliberate litter in terms of numbers of items is smoking-related materials. This is followed by fast-food/convenience packaging, which accounted for 20 per cent of all the items collected, but covered 25 per cent of the total littered area. In contrast, smoking materials covered only 5 per cent of the littered area. Beverage containers account for about 6 per cent

Table 3. Litter composition and accumulation

Category	Total count	Percentage of total	Percentage of littered area	Weekly accumulation ^a
Glass (total)	1025	10.3	2.3	137.5
Whole	79	0.8	1.5	5.7
Broken pieces	946	9.5	0.8	131.8
Metal (total)	732	7.3	2.3	75.0
Aluminium	657	6.6	2.1	64.2
Ferrous metal	3	0.0	0.0	0.5
Bi-metal cans	7	0.1	0.1	0.2
Wire	13	0.1	0.0	1.0
Miscellaneous	52	0.5	0.1	9.0
Paper (total)	2892	29.0	40.6	354.8
General	1005	10.0	11.5	121.5
Packaging	1887	18.9	29.1	233.2
Plastic (total)	1414	14.2	40.8	176.3
General	419	4.2	5.6	48.8
Packaging	995	10.0	35.2	127.5
Polystyrene	783	7.8	4.6	103.5
Smoking (total)	2810	28.1	4.9	345.8
Cigarette butts	2321	23.2	0.5	281.0
Other	489	4.9	4.4	64.8
Miscellaneous	333	3.3	4.5	56.8
Total	9989	100.0	100.0	1249.5

^a Number of items per week

of items littered and constitute only 3 per cent of the littered surface. However, New Jersey does not have a beverage container law, but does have a mandatory recycling programme for newspapers, glass, aluminium and yard waste. So while pervasive in number, the visual impact of smoking-related litter is less than fast-food packaging.

Accumulation rates

Accumulation rates are affected by many factors including the behaviour of people who litter, clean-up programmes, and season. During the survey, litter was not picked up by road or maintenance crews at any of the sites. Accumulation rates were calculated based on the amount of litter collected on the second and third visits and then divided by 4 (the total number of weeks sampled) to get an average weekly rate of accumulation. In all other litter surveys, accumulation rates were simply determined by one return visit. A wide range of accumulation rates for different types of litter was found (Table 3). Cigarette butts were the fastest-accumulating litter item, averaging about 281 cigarette butts per week over all sites combined. Paper packaging accumulated at the next-fastest rate, averaging 233 new items per week. Finally, broken glass appeared at a rate of about 130 fresh clusters per week for the total sample.

Fast accumulation rates do not translate into great visual impacts. For example, smoking-related items appear at a rate of 346 per week, compared with 355 for paper items. Yet paper accounts for nearly 41 per cent of the freshly littered area, while smoking materials cover only 2 per cent. Plastic accumulates by 176 pieces per week on average (half as quickly as paper), yet has about the same amount of visual impact as paper (40 per cent of total freshly littered area). It also persists longer in the environment if not removed. Clearly, the accumulation rate is important, but equally important is the amount of surface covered by litter. The source of the littered material is also important. It was found that 64 per cent of the freshly littered area consisted of packaging materials made from either plastic or paper. Paper packaging for snacks (such as gum and sweets) accumulates at the rate of 106 items per week, the highest rate of all the subcategories. The number of littered items, their size, and their persistence in the environment are all important in determining the environmental impact of litter.

Litter and land use

One of the most important goals of this survey was to determine similarities and differences among a variety of land-use categories in litter composition and accumulation rates. By dividing the sites into several land-use categories (Table 4) the importance of land use as a factor influencing both litter composition and accumulation rates can be seen. The use of geographical techniques in sampling methods (site selection) and data collection (GIS and standardization of grid size) facilitates such comparisons.

Commercial sites are the most littered type of land use, since 36 per cent of all litter counted was found in these locations. When standardized by the number of different sites in each category, commercial areas still have the highest number of littered items per site (446). This is followed by state parks (304 items per site), roads (262 items per site), landfills (257 items per site), beaches (234 items per site), industrial areas (191 items per site), residential areas (189 items per site) and watercourses (156 items per site).

Table 4. Land use and accumulation rates

Land use	No. sites	Total items	Percentage of total	Percentage of littered area	Weekly rate per site ^a
Residential	10	1891	18.9	39.4	21.1
Rural	2	222	2.2	5.1	9.4
Suburban	5	364	3.6	7.8	8.1
Urban	3	1305	13.1	26.5	50.7
Commercial	8	3574	35.8	68.8	51.1
Rural	2	246	2.5	6.3	16.4
Suburban	4	1474	14.8	20.7	43.5
Urban	2	1854	18.6	41.8	101.0
Industrial	3	574	5.7	13.7	26.7
Roads	6	1577	15.8	26.7	36.6
Beaches	4	939	9.4	14.4	35.4
Waterways	2	311	3.1	3.2	22.0
State parks	2	608	6.1	8.9	38.9
Landfills	2	514	5.1	10.8	33.3

^a Number of items accumulated per week per site

Deliberate litter is most prevalent at road, beach, water course, and commercial sites, while smoking-related items are found at all sites (Fig. 2), accounting for more than a third of all the items found in commercial sites, roadside sites, water courses and beaches, although it is less important in terms of total area littered. In residential sites, fast-food packaging accounts for nearly 25 per cent of the littered items and 22 per cent of the area littered. Beverage containers are not a major component of litter in any site (Fig. 2). In a breakdown by level of urbanization, smoking materials are the predominant litter type, by item count, in suburban residential and commercial locales. In both area and quantity, fast-food packaging represents the most pervasive litter problem in urban residential and urban commercial areas. Finally, the greatest amount of beverage container litter is at suburban commercial sites (about 10 per cent of total litter by item) and commercial sites in urban areas (9 per cent of total litter by item).

Community differences

As mentioned earlier, 21 communities were differentiated on the basis of their residential, commercial and industrial character and further subdivided into rural, urban and suburban categories. In addition, 16 communities representing more specialized environments were included in the survey. Urban areas are the most littered, accounting for 32 per cent (3159 items) of the total litter counted. Suburban areas are next (18 per cent or 1838 items), followed by rural areas (5 per cent or 468 items).

Residential sites. Residential sites ($n = 10$) accumulate fresh litter at an average rate of slightly more than 21 items per week (see Table 4). This figure varies widely, depending upon where the residential site is located. Urban residential sites, for example, each accumulate an average of nearly 51 littered items per week, while suburban and rural residential each sites accumulate fewer than ten items per week. There are also

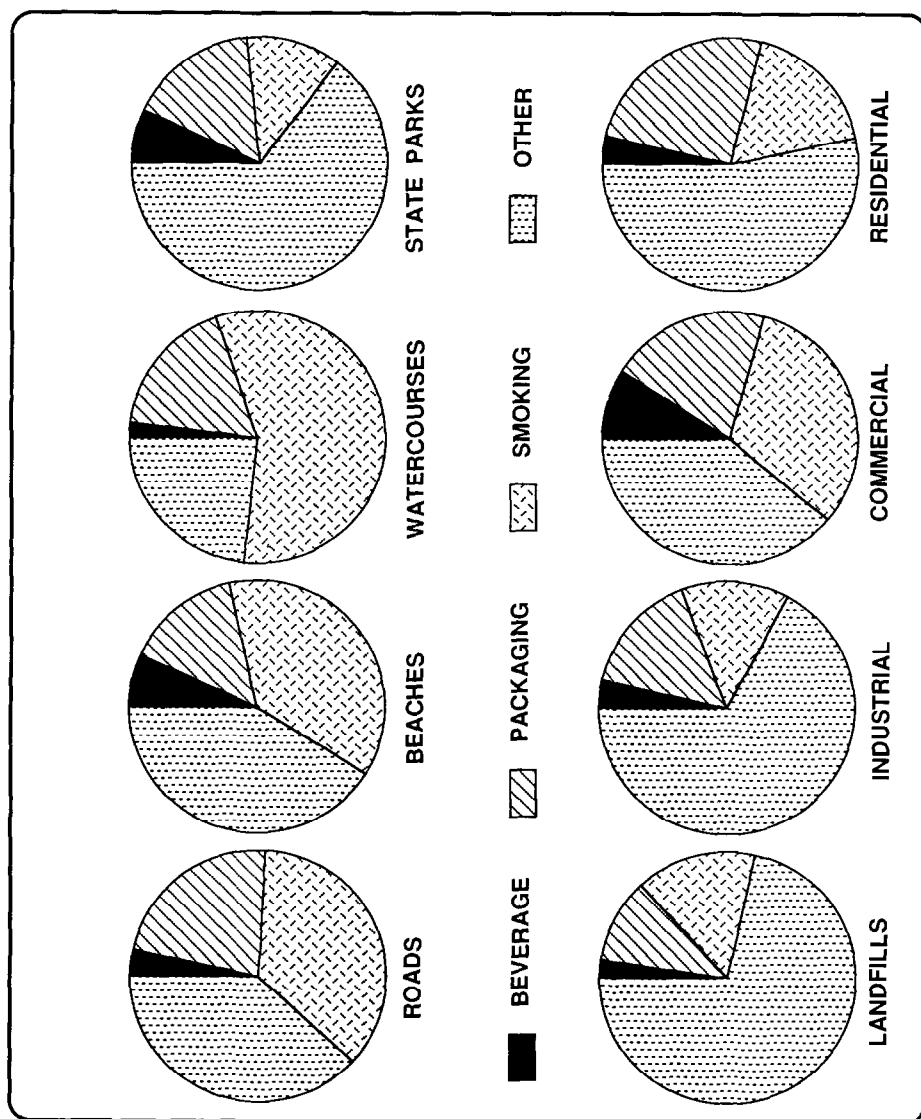


Figure 2. A comparison between the source of litter and a variety of land uses (by item count).

some differences among urban residential sites. There is less variation among suburban residential areas and among rural residential areas. By source, 22 per cent of the littered area in residential areas is covered with fast-food/convenience packaging. Convenience packaging litter has the widest coverage in urban residential areas. Smoking materials, by area, are more dominant in suburban residential areas than in urban or rural ones. On the other hand, beverage containers, by area littered, are more prevalent in rural and suburban residential communities.

Commercial sites. The eight commercial sites in the sample accumulate fresh litter at a rate of nearly 51 new items per week. The figure varies widely depending on the location of the site. For example, commercial sites in urban areas have an accumulation rate of 101 fresh pieces of litter per week. This contrasts sharply with commercial sites in rural areas, which have accumulation rates of the order of 16 fresh items per week. Suburban commercial sites (generally strip developments or shopping malls) accumulate fresh litter at a rate of almost 44 new items per week (see Table 4), again with significant variations between sites.

By category, fast-food convenience packaging is the most pervasive item of deliberate litter in urban commercial sites, while smoking materials are most prevalent in suburban and rural commercial areas. Fast-food packaging also covers the most littered area, ranging from 16 per cent in rural commercial sites, to 20 per cent in suburban commercial sites and 34 per cent in urban commercial areas. Despite their small size, the sheer volume of smoking-related litter poses significant problems in suburban and rural commercial locations, not only in total numbers but in the area covered. These items account for 9 per cent of the littered surface in suburban commercial areas and 7 per cent in rural commercial locations.

Industrial sites. The three industrial sites in the survey accumulated litter at an average weekly rate of about 27 new items per site (see Table 4). This is slightly higher than the rate at which residential sites accumulated fresh litter. There is a wide variation among industrial sites, primarily explained by density of overall development. Generally, less litter is found in rural industrial locales than in urban industrial ones. Most of the litter consists of paper pieces of indeterminate origin, styrofoam packing materials, and styrofoam pieces of largely unknown origins although the prime suspects are coffee cups, fast-food containers and coolers. Fast-food packaging, for example, comprises 15 per cent of the total items littered and covers 19 per cent of the littered surface.

Special environments

Specialized sites (roads, beaches, landfills) have remarkably similar rates of litter accumulation (see Table 4). Most litter is found on or along roadways (1577 items). However, when standardized by weekly rate per site, fresh litter was found to accumulate at a rate of 37 items per week on roadways. This is similar to the rate of fresh-litter accumulation in other public areas such as beaches (35 items per week), landfills (33 items per week) and watercourses (22 items per week). State park sites have the highest weekly accumulation rate (39 items per week), surpassing both industrial and residential sites.

By item count, smoking-related items are the most prevalent, numbering more than 57 per cent of the items counted for watercourses, 37 per cent for beaches, and 36 per cent for roads (Fig. 2). Due to their small size, their areal coverage is equally small (1–7 per cent of littered area). Areally, the most pervasive type of deliberate litter in

the special environments is fast-food/convenience packaging. Very little of the littered area (less than 5 per cent), however, is covered by beverage containers.

The importance of context in understanding litter rates

One of the important issues in understanding the nature of litter accumulation is the context within which it occurs, whether it is at the beach, along a rural roadway, or at a suburban shopping mall parking lot. Thus, land use, neighbourhood income, housing density and the local microclimate can influence the rate of litter accumulation at a specific site.

There are some differences in the quantity and rate of litter accumulation between different site types (urban, rural, suburban, residential, commercial, industrial and specialized environments) in this survey. In order to determine whether the quantity and rate of litter accumulation was simply a function of the number of sample sites in each land-use category, or indicative of true differences in litter accumulation, a Mann-Whitney rank, non-parametric statistic was computed for each land-use category for each major litter source. Despite different numbers of sample sites in each land-use category (urban = 5, suburban = 9, rural = 4), there is no statistical difference in total item count, composition or weekly accumulation rate between suburban and rural locales. There are, however, major differences in item counts, composition and weekly accumulation rates when comparing urban and suburban sites, and rural and urban ones. There simply is more litter generated in urban locations. Differences also occur in the composition of the litter between urban and suburban locales and between urban and rural locations (Table 5). Glass, paper, plastic and polystyrene (styrofoam) are found more often in urban places than in suburban and rural ones.

In addition to location, another important factor in understanding the generation and impact of litter is who is most affected by litter accumulation. In order to answer some of these questions, the total number of litter items collected and the weekly accumulation rate for each sample site were analysed and compared to some general statistics regarding the nature of the community in which the sites were located. While there is a danger in drawing any steadfast conclusions about the relationship between the specific sampling site and the community as a whole, some interesting findings did emerge. For example, there is no relationship between total litter count (number of items per community) and population density or road density (road miles per square mile) in the sample (Table 6). There is also no correlation between car and pedestrian

Table 5. Density comparisons by litter type (Z scores for Mann-Whitney statistic)

	Urban/suburban	Suburban/rural	Urban/rural
Glass	-3.50 ^b	-0.22	-3.39 ^b
Metal	-1.71	-1.04	-1.45
Paper	-4.65 ^b	-1.07	-4.72 ^b
Plastic	-2.88 ^b	-0.29	-2.64 ^b
Styrofoam	-2.44 ^a	-1.32	-2.16 ^a
Tobacco-related	-1.03	-1.02	-0.94
Miscellaneous	-0.97	-0.04	-0.94
Total	-3.39 ^b	-1.18	-3.01 ^b

^a $p < 0.05$

^b $p < 0.001$

Table 6. Relationship between total litter, weekly accumulation and selected community characteristics (Pearson's *r*)

Variable	Total litter	Weekly accumulation
Road density	0.19	0.14
Clean communities (nom.)	-0.04	-0.05
Recycling tonnage	0.20	0.15
Per capita income (\$)	-0.21	-0.22
Percentage African-American	0.45 ^a	0.40 ^a
Percentage Spanish-speaking	0.08	0.08
Population density	0.09	0.09
Percentage vacant land use	-0.10	-0.11
Percentage single-family land use	-0.44 ^a	-0.41 ^a
Percentage farm land use	-0.17	-0.17
Percentage commercial land use	0.77 ^b	0.71 ^b
Percentage industrial land use	-0.08	-0.09
Percentage apartment land use	-0.02	-0.02
Cloud cover (nom.)	0.14	0.16
Temperature	-0.12	-0.12
Wind speed	-0.09	-0.13
Pedestrian traffic (ord.)	-0.01	-0.02
Vehicular traffic (ord.)	0.14	0.12
Precipitation (nom.)	-0.09	-0.07

^a $p < 0.05$ ^b $p < 0.001$

Note: Unless otherwise indicated (nom. = nominal, ord. = ordinal), all variables are interval level in measurement

traffic at the sample site and total litter counted. These findings also relate to the weekly rate of litter accumulation. In addition, there is no apparent association between per capita income level in a community and total litter count and weekly accumulation. Finally, there is no association between weather variables (cloud cover, wind speed, precipitation) and litter accumulation.

The only three variables examined which demonstrate any type of relationship to total litter count and accumulation rates are the percentage of commercial land, the percentage of single-family residential land, and the percentage African-American population. The higher the proportion of commercial land use in a community, the more litter is accumulated. There is also a mild relationship between total litter count and weekly accumulation rates and race: the higher the percentage of African-American residents in a community, the greater the likelihood of higher litter accumulation rates. This can be partly explained by these areas having fewer community services, resulting in poor litter pick-up, and the lack of litter-reduction educational efforts.

Finally, the recycling effort in a community and participation in the Clean Communities programme (a state-sponsored grant programme targeted at litter reduction) have very little association with total litter count and accumulation rates. This may be explained by the newness of the Clean Communities programme, which first initiated its grant programme in 1988.

Conclusions

A number of conclusions from this survey highlight the usefulness of the approach and the role that geography can play in providing cogent data for environmental managers. The rate of fresh litter accumulation averages about 1250 pieces per week during summer for the entire sample. This is not a particularly voluminous amount, suggesting that litter is often cleared and re-accumulates quickly. The composition of litter is dominated by paper and plastic packaging when measured by area covered, and by cigarette butts when measured by piece count. There is some variation in the rate of fresh litter accumulation by community type. Urban areas (residential and commercial sites) generally have the highest rate of fresh litter accumulation. Commercial and industrial sites have higher litter accumulation rates than residential areas. Deliberate litter based on the source of the item (fast-food packaging, beverage containers, smoking materials) makes up an estimated 54 per cent of the total items counted and 33 per cent of the total area littered in our sample.

One of the largest failings of previous litter surveys is the inability to compare the results from state to state because of differences in site selection, site size, sampling methodology and frequency of sampling. Although 17 states have conducted litter surveys, it is virtually impossible to compare the results and thus the effectiveness of policy alternatives such as litter reduction programmes, bottle bills, and educational efforts. The need for a standardized methodology that permits comparisons between states and within states is crucial for our understanding of litter and its management.

The results of this survey cannot be used to calculate an overall estimate of the weight of materials littered in the state. They can, however, provide baseline conditions for each of the 37 sampling sites and be used to determine conditions in different types of communities within the state. Using the specialized environments sample for baseline conditions is more problematic, as they are less representative of the totality of all beaches, state parks and watercourses. It is virtually impossible to measure each piece of litter counted in the field. As a result, the estimates of areal coverage should be viewed with caution. Finally, the use of this technique (visual and full-scale survey) with a GIS approach is a significant improvement over the existing work in the field. If this methodology were to be adopted by other states, then comparisons between states could be made.

The application of this work to other arenas is important. Instead of extensive (and costly) statewide surveys, specific areas or communities should be targeted for more intensive analyses of litter. These mini-surveys (certainly utilizing some of the methodology presented here) could provide local data for the municipality. Having this baseline information would help the communities and the state to determine the effectiveness of any litter reduction effort, particularly if pre-programme and post-programme surveys were available.

There are a number of interesting policy questions which derive from this analysis. For example, based on the sample's compositional analysis, soft drink and beverage container legislation ('bottle bills') would have little effect on the quantity and areal coverage of litter in New Jersey. Efforts aimed at packaging (both paper and plastic), polystyrene (styrofoam) and tobacco products would have a far greater impact on litter reduction. Filters on cigarettes which biodegrade quickly would be one obvious suggestion. Another would be to target education efforts to highlight the fact that cigarettes are litter and do not simply disappear.

Finally, this research illustrates the role that geographers can play in environmental management. The range of geographical techniques lends itself to environmental problem solving at a variety of spatial scales. Litter is another facet of the solid waste

management issue in which geographers can have a significant impact and utilize their skills in a public policy context.

Acknowledgements

We wish to thank J. K. Mitchell and an anonymous reviewer for their helpful comments on this paper. Funding for this project was provided by the Office of Recycling, Division of Solid Waste Management, NJDEP, Grant No. 89-1. We gratefully acknowledge their support.

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(Revised manuscript received 20 November 1990)